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Bachelor Thesis

Do Shadows help in Damage Detection using Deep Learning Models?

Author: Sakshi Sharma (2761339)

1st supervisor: Bernd Andeweg

2nd reader: Wan Fokkink

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1 Abstract

Earthquakes usually occur when the shifting of tectonic plates within the earth surface takes place, releasing some powerful energy and causing huge damage and destruction across the planet. However, a quick and rapid damage assessment remains a challenge. Therefore, this thesis explored a new approach to detect earthquake-induced damage using satellite imagery and deep learning models augmented with shadow-based features to detect and classify damage caused after an earthquake.

The Mexico subset of the Multi-Earthquake Damage Building Footprint Segmentation (MEDBFS) dataset [1] has been used for this research. It consists of 1,048 images, each with high-resolution tiles (0.5 m) captured before and after the Mexico Earthquake on September 19th, 2017. The data set consists of pre- and post-disaster satellite images along with ground truth damage masks, which have been labeled into Intact, Severe and Collapsed type. The data set also provides some extra features in five dimensions, such as spectral, texture, edge detection, building index, and temporal sequencing, resulting in a total of sixteen distinct classes of feature images.

The research methodology involved a 10-channel input tensor per tile by stacking pre- and post-disaster RGB images (6 channels), MBI (1 channel), pre- and post-disaster shadow masks (2 channels), and a binary shadow difference map (1 channel). The shadow difference was computed from pre and post event using Hue, Saturation, and Value (HSV) based shadow masks. These inputs were then used to train a deep learning semantic segmentation model (U-Net) for pixel-wise damage classification and then compared to ground-truth labels.

This model was then compared to the baseline model, which had only 2 channels containing grayscale pre- and post- disaster images, and was also trained on a simplified U-Net model. The result showed that the 10-channel U-Net model outperformed the baseline model, particularly in metrics like Accuracy, F1 and Intersection over Union (IoU) in producing predictions of damage very close to the ground-truth labels. The model did have limitations, such as being unable to predict collapsed pixels at all and was often misclassified as severe damage. In general, the results presented that shadow-informed deep learning models have the potential for a more accurate assessment of disaster damage. But, in order to apply the models developed in this research paper to real-world settings, further research is needed.

2 Introduction

2.1 Context

This research focused on the Puebla-Morelos earthquake [2][3] that had hit Mexico on September 19, 2017 with a 7.1 magnitude. The damage caused led to 370 confirmed fatalities and many building collapses in just Mexico City. The epicenter [1] of this quake was located in the state of Morelos which is about 120 km from Mexico city. The Puebla earthquake occurred just twelve days after an earthquake of 8.2 magnitude had hit the coast of Chiapas(close to Mexico city) on September 7, 2017 making the seismic condition worse there.

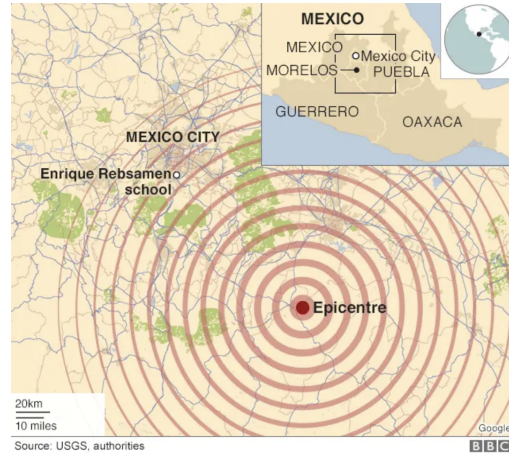


Figure 1: Puebla Earthquake Epicenter [4]

However, some studies showed that the 2017 Puebla earthquake [3] was very different in nature compared to the major earthquakes that had hit Mexico. Earthquakes in Mexico usually occur along the Pacific coast due to interaction between the Cocos and North American plates. But the Puebla earthquake strangely originated in the interior of the country far from its seismic zone. This made the Puebla earthquake a very rare event. In the past, only five such inland earthquakes have been reported worldwide[2].

To make matters worse, the Mexican city is built on a dry lake bed [2] that has been filled with water-related sediments such as soft clay, silt, and sand instead of solid rock. This geological structure in Mexico makes the tremors much stronger and more dangerous than in the surrounding areas. This happens because the watery sediments slow the seismic waves and the energy they contain is changed into severe shaking. Due to this, earthquakes that strike farther away also cause massive damage in the capital. Mexico City has a population of 21 million, which leads to a very high number of casualties due to unstable ground[3].

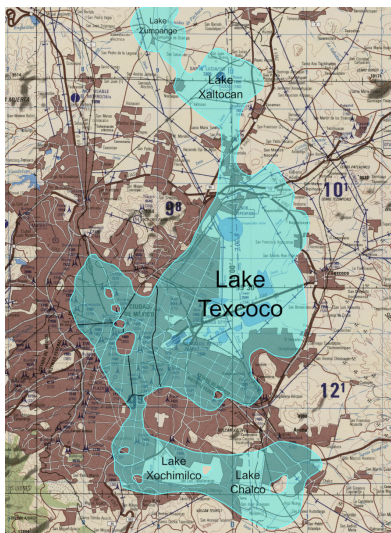


Figure 2: Mexico built on ancient lake bed [5]

Around 184,000 homes and 16,000 schools were reported to have been damaged or destroyed in the Puebla and Mexican regions [3]. The Mexican states of Puebla and Morelos, along with Mexico City, had faced intense pressure of destruction as a result of the 2017 Puebla earthquake[6].

2.2 Research Gap and Problem Statement

The Puebla earthquake till date stands a strong proof of how risky earthquakes were and still can be for Mexico due to its unstable ground. What is very concerning is that even though Mexico is one of the most vulnerable countries on Earth today to seismic events, it has not yet taken any steps to prepare for a disaster. While Japan also faces frequent earthquakes, but are always prepared to face them. They have spent a lot of money on early warning systems, public education about safety during earthquakes, and earthquake-resistant buildings. They also make use of advanced technology to assess damage as soon as possible. This preparation and quickness helps in damage control[7][8].

This research has attempted to address the existing gap in damage assessment by using a novel approach that is based on shadow analysis through satellite images. Methods used in the past to assess damage are considered to be slow, time-consuming, and limited in scope. Satellite images, instead, provide a faster way to monitor large areas of damage. Some areas after an earthquake become really inaccessible due to infrastructure collapses, and that is where satellite images can be useful. When these satellite images are combined with deep learning models like U-Net, shadow features extracted from satellite images could provide a better and faster damage assessment.

This could be of great help for the damage response teams as they can get important and quick information from the models about the kinds of damage and the regions affected after an earthquake. These models can also help Mexico being better prepared for facing earthquakes in future as it could suggest ways to improve the infrastructure by detecting which type of buildings are more likely to collapse.

2.3 Research Objectives and Questions

The study aims to investigate, if shadow features from satellite images are capable of detecting structural damage using deep learning models by its particular focus on the 2017 Puebla earthquake in Mexico [3].

The research questions include:

- How well can the use of multi-input shadow channels help in damage detection?
- Can deep learning models be trained to classify earthquake-damaged buildings in Mexico using shadow features?
- What are the main limitations of using shadow analysis along with deep learning model for earthquake damage detection in satellite imagery?
- Can shadow based features effectively contribute to real-time emergency response and guide long-term good infrastructure planning?

These sub-questions lead to the rise of the main research question:

- **Can shadows in satellite image be detected and extracted to help in detecting damage caused by an earthquake using an U-NET model?**

3 Literature Review

Damage detection in past after an earthquake has always been the most difficult task in urban areas because of how densely they are populated and sometimes have really poor infrastructure. Over the past two decades, significant progress has been made across remote sensing, computer vision, shadow modeling, disaster response using AI, and many more. This section reviews existing literature across various domains related to this study.

3.1 Damage Assessment in the past

Damage assessment in 1990s relied completely on manual efforts such as field visits to the affected area, interviews with the locals, and affected people. These efforts used to be documented in reconnaissance reports and were very time consuming and limited in scope. One such study, reviews twenty-nine reconnaissance reports from developing countries, between 1990 and 1998, identifying that most structural damage was due to lack of proper structural design, poor detailing and construction malpractices[9].

Their work basically showed the vulnerability of buildings built without proper engineering tools or strong materials and highlighted the immediate need of stronger building codes, local training, and better construction oversight. However, these early assessments lacked scalability and real-time applicability, which modern techniques now aim to address[9].

3.2 Emergence of Remote Sensing in Damage Detection

The emergence of remote sensing technologies are increasingly used in structural damage detection due to its key advantages like wide-area coverage, no contact data collection, low cost and lastly faster response. These features make them very useful for monitoring large-scale damage in real time. A wide range of remote sensing data sources have been introduced like radar data, LiDAR and many more have been employed for detecting damages in both buildings and bridges. They help to explore and improve detection accuracy and efficiency by overcoming the limitations of traditional field surveys and gained valuable insights in real time, which is really important for effective disaster response and recovery[10].

3.2.1 Why is satellite imagery reliable for damage detection

The published work [11] investigates the 2017 Sarpol-e-Zahab earthquake in Iran case study and sees how satellite imagery(a remote sensing technique) can be used to detect earthquake damage. The earthquake, had hit Iran with a magnitude of 7.3 and caused a lot of damage to the infrastructure and people.

The researchers used Sentinel-2 satellite images, consisting of pre- and post-disaster earthquake imagery. The method involved two machine learning models, Support Vector Machine (SVM) and Neural Networks (NN) to classify the damage into "change" or "no change" and then compared the results using Google Earth and field reports.

The research used medium-resolution satellite imagery because of their clean quality and up-to-date view of large areas that simplified the research process. Hence, the published paper considered satellite images to be very reliable.[11].

3.2.2 U-NET for semantic segmentation in remote sensing

The U-Net architecture, was first introduced by Ronneberger et al.[12] which is a convolutional neural network specifically for biomedical image segmentation. The architecture is well known for its U-shaped structure consisting of a downsampling path(encoder) which captures important features by reducing image size and an expansive path (decoder), also called an upsampling path which helps to increase image size back to original and then combines the details from the encoder enabling precise segmentations.

Modern U-Net implementations presented by TensorFlow and Keras [13] have made the architecture more accessible and adaptable. Along with that, GeeksforGeeks [14] provides a clear theoretical overview of U-Net.

Using the previously mentioned existing architectures and insights, the model presented in this research is capable of handling multispectral image data. Unlike standard U-Net which typically accepts RGB(3-channels) or grayscale(1-channel) with few channels, but the model in this paper is capable of handling 10 channel input, making it suitable for tasks such as remote sensing or satellite imagery analysis. Some more major changes compared to the other existing model are that UpSampling2D has been used instead of transposed convolutions (Conv2DTranspose) for the decoder to reduce computational complexity. The architecture maintained the core U-Net structure that is downsampling blocks, a bottleneck, three upsampling blocks but is implemented using Tensor Flow Keras API to simplify the process of experimentation[14] [15].

The output layer uses a softmax activation[13] to support multi-class segmentation with number of output channels with three classes [0: Intact, 1: Severe, 2: Collapsed]. These simplifications and improvements helped to improve the effectiveness of the original U-Net model.

So the main reason why U-Net has been chosen for this study, is because it fits the image segmentation tasks very well, especially with respect to satellite images. Its encoder-decoder structure along with skip connections helps to preserve spatial information required to understand and assess building damage. The model is capable enough to handle multi-input channel and also perform multiclass segmentation, which makes the U-Net model a good fit for this research.

3.3 Role of Shadows in earthquake damage detection

The researchers of the published work believe that shadows play a very important role in identifying earthquake-induced building damage with high-resolution satellite imagery. The published paper demonstrated that the shape, structure, and intensity of shadows can offer hints about the structural integrity of buildings after an earthquake. The research believes that shadows can be extracted using two methods which are Property-Based Methods and Model-Based Methods. Property-based extraction method means to obtain shadows by using specific properties like texture, color, shape. While, model-based method means extracting shadows according to the relationship between buildings and their casted shadows[16].

In the published paper, the researchers used both methods, that is, constructed theoretical shadow polygons from pre-disaster earthquake building footprint height data (model-based) and then analyzed pixel-level brightness values with actual shadow regions. This helped them identify and classify buildings that were fully collapsed, partially collapsed and even undamaged[16].

However, what makes shadows a good feature for damage detection is that any subtle deviations in building shape can show a difference in casted shadow, such as short cast or no shadow at all, leading to a good damage assessment.

The results of the published article motivated the choice of using Property-Based Methods for the present study [16].

3.3.1 Shadow Detection HSV Method

The choice of HSV color space in this published paper is motivated due to its ability to separate the intensity component (V) from the chromatic components (H and S), which is very advantageous for shadow detection. Shadows, in general, result in a drop in brightness(V), meaning when a shadow falls on something, it makes the area dark but does not change the color or how strong the color looks. In the HSV space, this means that the Hue (color type) and saturation (color strength) stay the same almost, while the value (brightness) gets lower. This separation allows shadow detection to focus mainly on changes in brightness rather than being confused by color differences. Also, when saturation is very low (like grayish), only the brightness component is considered, which helps to avoid mistakes when detecting shadows on low-color areas. Hence, HSV color space does provide an effective way to detect shadows by isolating brightness changes from color information mainly during complex lighting conditions. The findings of the published paper motivates the reasoning of using HSV shadow method for the present study[17].

3.4 Importance of Multi-Input Channel for Damage Detection

The paper highlights that the use of multichannel features allows shadow detection methods to capture extra and important information from different aspects of the image. Although individual color channels like RGB provide basic color information along with thresholding techniques on intensity and brightness channels, but it still helped to identify the shadow regions by isolating areas with lower illuminations. Hence, it can be said that multi-input channels with thresholding helped to improve the accuracy and efficiency of shadow detection in complex lighting conditions[18].

Inspired by this, the current research used the multichannel approach by combining different inputs along with adaptive thresholding to assess and improve earthquake-induced shadow detection in satellite imagery.

3.5 Gaps

Although many advances have been made with respect to earthquake damage detection using satellite imagery, current studies have focused on individual aspects such as shadow modeling, multi-channel feature extraction, or deep learning segmentation separately. No or very few research has integrated all these components, that is, combining satellite pre- and post- disaster images, multi-input channels along with thresholding techniques for shadow detection and deep learning model like U-Net into a complete framework for assessing the damage caused by earthquake. Hence, this gap motivates the current study to explore the advantages and disadvantages of integrating all these methods for better damage assessment.

4 Dataset

4.1 About

This paper focuses on MEDBFS [1], which is a large-scale data set to analyze earthquake-related damage using satellite imagery. It encourages the development of machine learning models for damage detection and classification, offering a wide range of well-annotated remote sensing data.

It consists of satellite images of five major global earthquakes such as Yushu (China, 2010), Puebla (Mexico, 2017), Turkey (2023), Morocco (2023) and Afghanistan (2023)[3]. It has around 7,000 pre- and post-disaster satellite images with standardized dimensions of 512x512 pixels which have been obtained from platforms such as QuickBird, WorldView, GF-2[1].

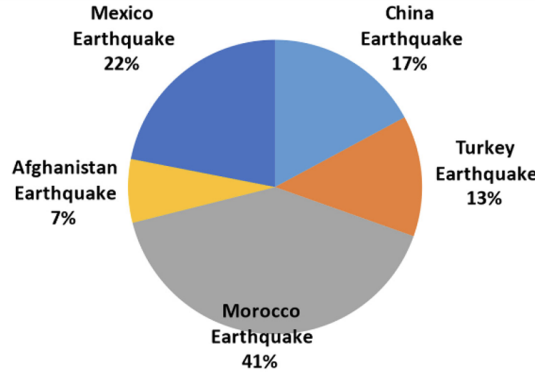


Figure 3: Datasets Piechart from MEDBFS [1]

The country data set used in this research is Mexico. The most critical work involved the classification and annotation of building damage into three primary categories based on visual characteristics.

- Intact damaged
- Severely damaged
- Collapsed

This classification is obtained from the European Macro Seismic Scale (EMS98) and the four damage levels of the xBD dataset. However, the obtained three damage labels are considered to be ground truth annotations that were performed by laboratory researchers and then reviewed by professional staff to ensure validity and authenticity.

In addition to pre- and post- disaster images and their labels, the dataset also provides sixteen feature maps extracted across five dimension namely, spectral(RGB), texture (GLCM, LBP), edge detection (Prewitt, Laplacian), building index (Morphological Building Index), and temporal spacing (3D GLCM), capturing various structural and visual indicator regarding the damage[1].

However, before feature extraction, raw data underwent extensive preprocessing steps such as quality and cloud cover screening, atmospheric correction, geographic matching, radiometric calibration, etc. The paper also proposes a feature tokenization network (FTN) to effectively extract visual representations from these inputs. Later, the data annotation protocol was validated using the FTN network training dataset to recognize collapsed buildings compared to the annotated labels[1].

The MEDBFS Mexico dataset [4], used in this research, is publicly available on GitHub under a CC-BY-NC license and serves as a valuable resource for research purposes[1].

4.1.1 Advantages of the MEDBFS Mexico Dataset[1]

- Addresses a critical data gap: The paper directly responds to the limited availability of large publicly accessible datasets for the assessment of earthquake-damaged buildings. It combines

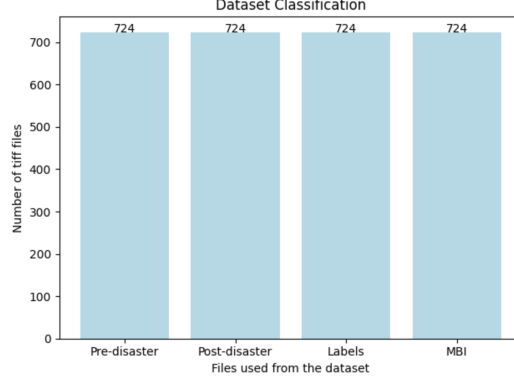


Figure 4: Dataset Classification

satellite images from multiple sources like QuickBird, Jilin-1, WorldView, Xbd, helping to avoid looking at multiple places for datasets.

- Multiple and rich features: This data set develops a comprehensive feature set that contains many features that help with a detailed research process.
- Data Preprocessing: The data have been very well preprocessed which indeed reduced the initial work of preprocessing for researchers like cloud and fog separation, orthorectification, geographic matching, and many more.
- Labeled ground truth: The best part of the data set is that it annotates the ground truth in order to help the models check their validity if the damages are recognized correctly or not.

4.1.2 Disadvantages of the MEDBFS Mexico Dataset[1]

- Lack of exact time and location metadata: Only the general date and information are provided, that is, the day the earthquake hit and the region. The exact timestamps (hour/minute) and precise image coordinates are not included and are not accessible publicly. This made the research process a little difficult as without this information, it was difficult to detect which part of Mexico was in the image and how much damage was caused in that region.
- Missing Meta-Data: The data set does not include the original sun or shadow angle data. This limits the accuracy of shadow-based damage detection and makes normalization a little difficult.
- Class Imbalance: The data set consists of the label folder that acts as a ground truth and consists of three damage classes: *Intact*, *Severe*, and *Collapsed*. However, the number of collapsed classes is much smaller compared to the other two classes in the data set that cause class imbalance [1].

Class	Description	Pixel Count	Percentage (%)
0	Intact	134,504,993	70.85%
1	Severe	55,197,802	29.09%
2	Collapsed	89,461	0.047%

Table 1: Class-wise Pixel Distribution and Percentage in the Dataset

5 Methodology

5.1 Environment and Tools Used

- Google Collab: I started the initial phase of coding and development using Google Collab due to its good accessibility and free access to GPUs and TPUs particularly aimed at deep machine learning tasks. During this phase, the focus was on developing foundational code, that is, handling data set access and preprocessing.

But as I continued to work on the thesis, Colab began to show some limitations, such as each training epoch taking more than three to four hours to complete and the available free disk space was also not enough to continue large-scale training. This led me to switch to a local environment.

- Google Drive: The data set was initially uploaded to Google Drive for Google Colab to easily access. This integration made data loading much easier and served as a central storage area at the beginning of development. Later, the data set was downloaded from the drive to the laptop desktop to be used by the local environment.
- Jupyter: Once I moved to the local environment, I used the Jupyter notebook for the training, testing, and visualization steps. The data set was then downloaded to the desktop and accessed by the Jupyter notebook. This approach drastically reduced the training time and allowed for easier debugging and visualizations with various Python tools.
- Terminal: The terminal was used to navigate between directories, install various dependencies, and even start helping to work with jupyter.
- GitHub: I maintained my project on GitHub for version control and organization. Making regular commits helps me track my progress and manage loads of code.
- Google Docs: Google Docs was used to plan the thesis workflow and write my draft version of the thesis report. I used to outline the key steps and track progress to help maintain the content of the final document.

5.2 Data Preprocessing

All images in the MEDBFS dataset had already undergone various rigorous preprocessing procedures, including quality screening, radiometric calibration, atmospheric correction, orthorectification, and expert validation to ensure consistency and accuracy across the dataset. These preprocessing steps were really important, as they provided a reliable foundation for training deep learning models without the need for extensive remote sensing corrections by the end user, for example, deleting images with cloud-covered tiles. Thus, the data set was appropriate for the initial phase of the research.

After obtaining the data set, two different input channels along with their unique preprocessing pipelines were developed and will be discussed below:

- A complex "10-channel multi-input channel"
- A simpler "2-channel grayscale input channel"

Following data set acquisition, some additional and basic preprocessing steps were applied to convert raw satellite imagery into structured input that would be compatible with deep learning pipelines.

The basic goal was to create a multichannel input tensors (basically a multidimensional data array that acts as input for neural networks) that combined visual, structural and shadow-based features to improve the model’s ability to detect and classify building damage.

5.2.1 Preprocessing for 10-Channel Multi-Input Model

Each tile in the image, which was 512 x 512 pixels in size, was converted into a 10-channel input with dimensions (512, 512, 10), where each channel showcased a different feature needed to assess damage to the building later. The first six channels consisted of pre- and post- disaster RGB images (three channels each), in order to give information to the model to compare structural differences over time. The seventh channel consisted of the Morphological Building Index (MBI), which was very closely related to shadows because it is a spectral feature that helps to highlight the edges and roofs of the building. The main reason for having this feature was to help the model identify urban areas by comparing them with areas of vegetation or open fields.

For better damage assessment, the last few channels focused a lot more on deriving shadow information. For both pre- and post-disaster RGB images, shadow masks were created by converting them to HSV color space first and then setting a specific value for the brightness in the V channel of the HSV color space (thresholding the brightness (V) channel). Pixels with low brightness, that is, when V was less than 0.3, were marked as shadows. These shadow masks were then added as the eighth and ninth channels.

For the last channel, a shadow difference map was developed in order to show the changes in shadow coverage. The main purpose of this shadow difference channel was to highlight areas where the shadow could suddenly appear or disappear to indicate structural damage. The shadow difference [5] was computed using:

$$\text{shadow_difference} = |\text{shadow_post} - \text{shadow_pre}| \quad (1)$$

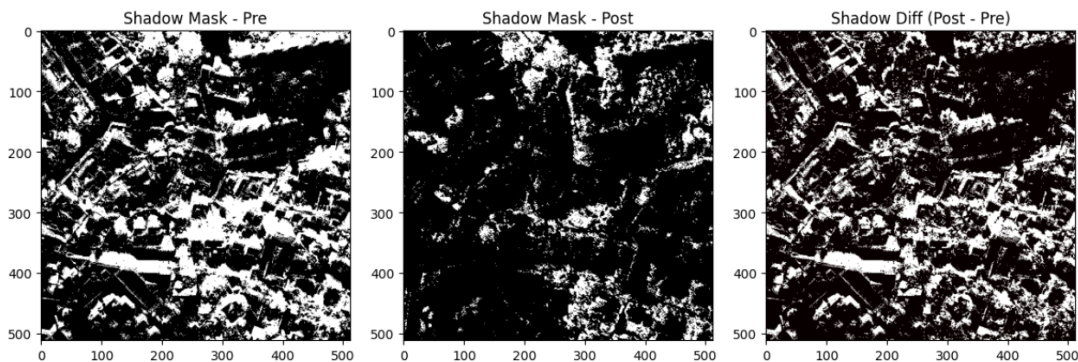


Figure 5: Shadow pre, Shadow post and Shadow difference

This was stored as the 10th channel, providing an additional signal for damage detection.

The purpose of the resulting 10-channel [6] [3] input model was to process both visual and structural hints and feed it into the U-NET model to detect damage accurately.

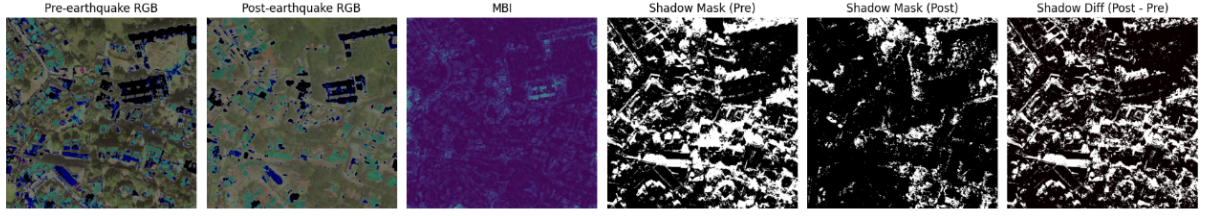


Figure 6: 10 Input Channel

Channel Index	Data Layer
0	Pre-disaster Green
1	Pre-disaster Blue
2	Pre-disaster Red
3	Post-disaster Green
4	Post-disaster Blue
5	Post-disaster Red
6	Morphological Building Index (MBI)
7	Pre-disaster Shadow Mask
8	Post-disaster Shadow Mask
9	Shadow Difference

Table 2: Channel Mapping for 10-Channel Input Tensor

5.2.2 Preprocessing for 2-Channel Multi-Input Model

For the baseline model, a very simple input channel was created by first converting the RGB pre- and post-RGB satellite images to grayscale.

For conversion, a very simple formula was applied, which is mentioned below:

$$\text{Gray} = 0.2989 \cdot R + 0.5870 \cdot G + 0.1140 \cdot B \quad (2)$$

Each RGB image was loaded and converted to grayscale images. These were normalized to $[0,1]$ and stacked to form a tensor $(512, 512, 2)$ [7].

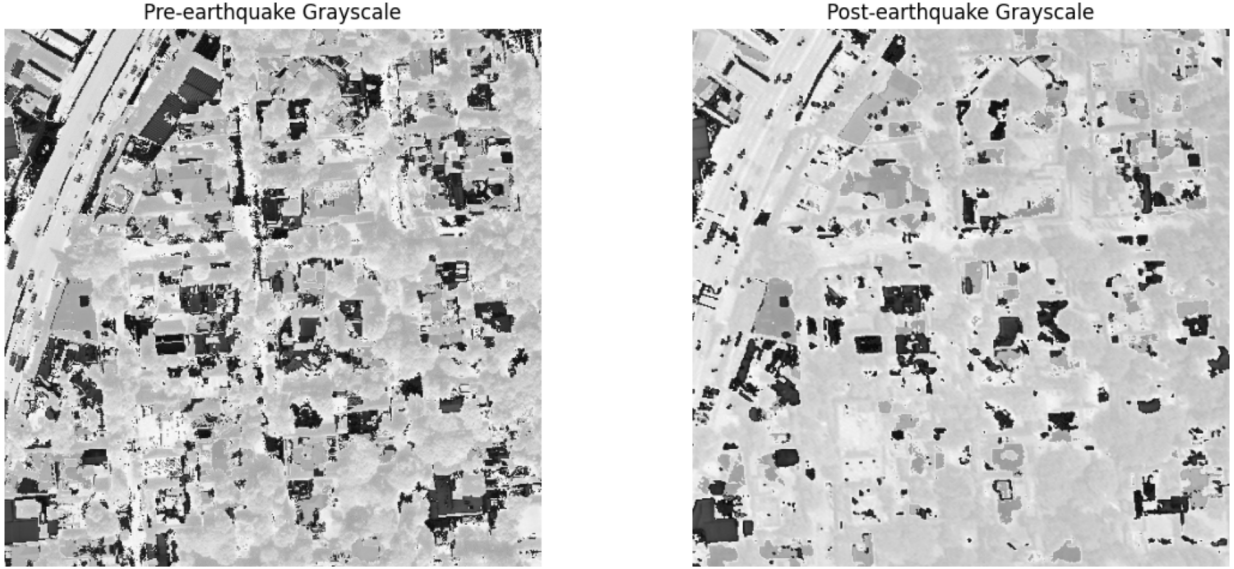


Figure 7: Grayscale 2-Channel Input

Channel Index	Data Layer
0	Pre-disaster Grayscale
1	Post-disaster Grayscale

Table 3: Channel Mapping for 2-Channel Input Tensor

The Label Masks in the dataset were already annotated with one of the three damage categories, that is, 0: Intact, 1: Severe, and 2: Collapsed, but in order to visualize them they had to be converted into grayscale images. However, the label masks were needed to be converted to uint8 format in order to match the input tile dimensions of the 512x512 pixels. Then these were considered ground truth for training and evaluation during supervised learning [8].

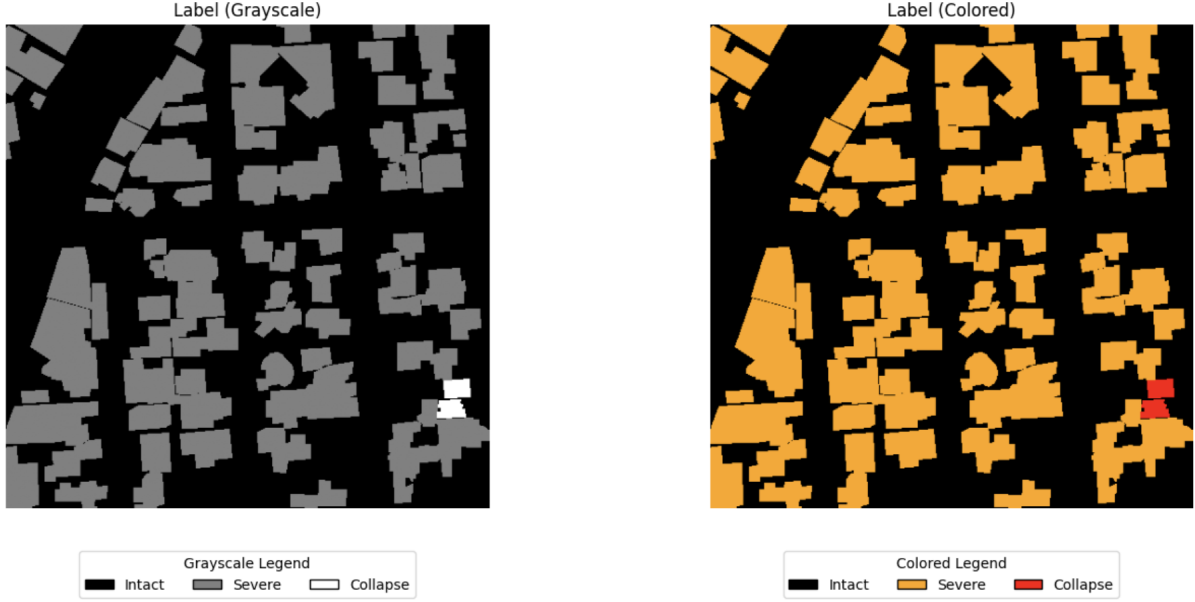


Figure 8: Annotated Ground-Truth Label Visualization

To efficiently access all the images during training and evaluation, all inputs and labels were stored as `.npy` files. This format made it easier to load the multichannel NumPy array and to be used consistently during training, testing, and validation. The final file structure consisted of input arrays, e.g. `part0cut0h0w.input.npy`, consisting of the 2 or 10 channel inputs paired with the corresponding label mask that is, `part0cut0h0w.label.tif`, all organized under the directory, `Processed_NPY_2ch_Grayscale` and `Processed_NPY` for the second or tenth channel respectively for efficient loading during training.

5.3 Model Architecture

For this research, two different U-Net-based models were used to perform semantic segmentation on satellite images of the earthquake-affected regions in Mexico. But both of these models are based on the classic U-Net architecture [14][12] which is a kind of neural network used for pixel-wise image segmentation tasks. It is widely used in the fields of medical imaging and remote sensing, where it is used to divide an image into different parts to identify and classify certain changes at the pixel level.

The U-Net architecture is symmetrical and has three main components:[15][19]

The Encode (Contracting Path): The first half of the model responsible for gradually shrinking the image (making it smaller) and then increasing the number of feature channels. This part helps the model learn simple and complex information, ranging from details about edges to more complex features like collapsed buildings.

Each encoder block has two convolutional layers with small filters (3x3) pixels needed to scan images and find features and is followed by an activation function called ReLU to add nonlinearity so that the model could learn better. It also has a 2x2 max pooling layer to reduce the image resolution[14].

Moving deeper into the encoder, the number of filters doubles at every stage to increase the abstract features. For example, the first block uses only 64 features with an output shape, then the second

uses 128, the third uses 256, eventually reaching 512 filters in the bottleneck area[14].

The number of feature channels doubles after each block[15]:

Block 1: 64 filters : output shape: (256, 256, 64)

Block 2: 128 filters : output shape: (128, 128, 128)

Block 3: 256 filters : output shape: (64, 64, 256)

Block 4: 512 filters : output shape: (32, 32, 512)

Bottleneck: This is the middle part of the model, which has the smallest image size but a lot of information about the features of the image. It does not shrink the image any further, but processes the smallest spatial resolution of the image using 1024 filters (since the image is too small, a higher number of filters are required to understand and extract information), capturing deep and complex of the image such as large shadow areas or sets of collapsed buildings. Therefore, this part is known to be the most compressed area where abstract information is stored.[14].

The Decoder (Expansive Path): This is the second half of the model, where it takes the deep, abstract features from the bottleneck part and slowly turns the images big again like to their original size using 2x2 unsampling layers. After each unsampling step, the decoder concatenates the unsampled features with the matching features of the encoder by skipping connections. This step is necessary because the encoder brings in very minute details, which the model should not miss out on as it could have been lost during downsampling. Each downcoder then applies 3x3 convolutional layers with ReLU activation to clean up and refine the features[15].

The number of filters halves in each decoder stage[15]:

Decoder 1: 512 filters : output shape: (64, 64, 512)

Decoder 2: 256 filters : output shape: (128, 128, 256)

Decoder 3: 128 filters : output shape: (256, 256, 128)

Decoder 4: 64 filters : output shape: (512, 512, 64)

The final layer of the model is a 1x1 convolutional layer with three output channels, one for each class label (0: intact, 1: severely damaged, 2: collapsed). In this layer a softmax activation function has been used to convert these values into probabilities[15]. This means that the model says how likely each pixel will belong to each damaged class. The class with the highest probability is chosen as the final prediction.

5.3.1 10-Channel Multi-Input U-Net Model

This model, as mentioned in the earlier sections, consisted of rich multi-channel input of shape (512 x 512 x 10). These ten channels included the pre-disaster RGB channel (3 channels), the post-disaster RGB channel (3 channels), MBI as a shadow proxy (1 channel), pre and post difference shadow masks (2 channels) and the calculated shadow difference map (1 channel). This approach was developed for the model to not focus on color and texture, but also looked at the structural and lighting information, particularly shadow-related hints in order to check it is possible to assess damage using shadows along with deep learning models. The U-Net model used here included all the components of a standard U-Net model. The output expected from the model was a segmentation map (type of image output by the model, in which each pixel is classified into a category) of shape (512, 512, 3) representing damage categories(0, 1, 2).

5.3.2 Simplified Model: 2-Channel Multi-Input U-Net Model

To compare the standard model, a simplified U-Net was developed using only two input channels. It consisted of grayscale pre- and post-disaster images with input shape (512 x 512 x 2).

In this version of the model, actually, the basic U-Net structure has been followed but with fewer filters and layers to simplify it.

1. Encoder: There were two encoder blocks with 16 and 32 filters and each filter consisted of 3x3 convolution layers with ReLU activation and 2x2 max pooling layer and dropout[15].
2. Bottleneck: This was the middle block, which had two 3x3 convolution layers and around 64 filters. These helped catch grayscale differences caused by damage[15].
3. Decoder: It had two upsampling 2x2 layers along with convolution layers with 32 and 16 filters. However, skip connections were not used here, but were used in the standard model[15].
4. Output: A final 1x1 convolution layer was used along with a softmax activation function to produce class probabilities across three damage categories. The output shape was (512, 512, 3)[15].

The main purpose of this model was to create a baseline to which it would be possible to compare the 10-Channel Multi-Input U-Net Model to see if shadows specific features helped in detecting damage.

5.3.3 Summary on the Models

Feature	10-Channel U-Net	2-Channel U-Net
Input Channels	RGB, MBI, shadow mask, shadow diff	Grayscale (pre and post)
Input Shape	(512, 512, 10)	(512, 512, 2)
Encoder Blocks	4 filters: 64 $\rightarrow$ 512	2 filters: 16 $\rightarrow$ 32
Bottleneck	1024 filters	64 filters
Decoder Blocks	4 filters: 512 $\rightarrow$ 64	2 filters: 32 $\rightarrow$ 16
Skip Connections	Yes	No
Output Shape	(512, 512, 3)	(512, 512, 3)
Input Capacity	High (multi-modal)	Basic (grayscale only)

Table 4: Summary of Differences between 10-Channel and 2-Channel U-Net

5.4 Model Splitting

The data set was split only once into 3 non-overlapping parts: 70% training, 15% validation, and 15% testing. The split was done to ensure that there were no tiles overlapping in the data set. The training set was used for model learning, validation set for tuning the model along with early stopping [20] in case of overfitting [21], and test set only for final evaluation. In this study, no cross-validation or fold-based splitting was used.

Sample files	Amount
Training samples	506 tiles (70%)
Testing samples	109 tiles (15%)
Validation samples	109 tiles (15%)

Table 5: Sample distribution

5.5 Model Training[22]

Function	About
Loss Function	Sparse Categorical Cross Entropy which is suitable for integer coded class labels
Optimizer:	Adam with a learning rate of 0.0001 (a default value that ensures a balance between fast convergence and training)
Epochs	10
Batch Size	4
Callbacks	1. EarlyStopping with <code>val_loss</code> as 5 and patience as 5 to stop training if the validation loss does not improve after 5 epochs 2. ModelCheckpoint which saves the best model based on the lowest validation loss

Table 6: Training Function Details

Adam Optimizer[23] was used to make the model run with atleast 10 epoch runs and then was trained using the SparseCategoricalCrossentropy[24] loss function which is the best used for multi-class classification problems where labels are integer coded, like 0,1,2 for damage classification. The primary performance metric used during training was SparseCategoricalAccuracy [25] which was then used to measure how closely the model’s prediction matched the ground truth at every epoch and then overall.

A total of 106,643 (416.57 KB) and 31,383,107 (119.72 MB), trainable parameters [13] were used in the 2-input channel and 10-channel U-Net, respectively. To prevent overfitting and ensure efficient training, EarlyStopping[20] (training halts if validation loss does not improve for 5 consecutive epochs) and ModelCheckpoints [26](saves the best-performing model on validation loss) callbacks were used.

5.6 Understanding how the model learns:

The learning process involved deep learning model like the U-Net which followed a supervised learning method where it was provided by ground-truth annotated labels to predict the results and then learn from its mistakes.

Each training sample consisted of either 512x512x10 (10 input channel) or 512x512x2 (2 input channel), where each channel in both inputs represented different features.

These input tensors are then fed into the model. The model then performs a forward pass (process of feeding input through network layers to produce an output/prediction)[27], where input is processed

through multiple layers such as convolutions, activations, pooling, up-sampling and many more to produce output predictions.

In the models, the output is in the form of 512x512x3 tensors representing the probability that each pixel of the image belongs to one of the three damage classes. The probabilities were found using SoftMax activation [28] in the final layer.

- **0:** Intact
- **1:** Severely Damaged
- **2:** Collapsed

How comparing predictions with ground labels worked?: The ground truth labels [29] for each input tile were a 512x512 matrix. It consisted of three integer values like 0, 1, 2 that represented damage classes. To evaluate how well the model was able to predict damage in comparison to what the ground truth labels showed, the output was therefore compared to ground truth labels using loss function (Sparse Categorical Cross-Entropy)[30].

Weight Calculation: After the loss was calculated, the next step was to calculate the loss gradient with respect to each model weight. This was done using backpropagation[27], where the model achieved the loss values by propagating backward through the network. These gradients were responsible for indicating how the weights should change to reduce the loss to match the ground-truth labels. However, to update the weight, the model used Adam Optimizer with a default rate of 0.001, where it made small adjustments to the weights to reduce the loss.

The entire cycle started from feeding the input to computing loss, backpropagation, and finally weight update, and this continued over 10 epoch runs.

The model never directly saw the labels during the prediction; it only learned from its mistakes. Ground-truth labels were always used after the prediction was made and when the loss was calculated. This made sure that the model was never able to memorize [31]. After every epoch, the performance was tested on the validation set. If the model stopped improving, the EarlyStopping callback stopped the training to avoid overfitting, and ModelCheckpoint was used to save the best model based on validation loss.

6 Evaluation and Results

The evaluation of the trained model was done using both quantitative metrics and qualitative metrics (visually). After training was completed, the model was tested the test set with 109 image tiles, which was kept entirely separate from both the train and the validation set to see how the model would work in unseen earthquake scenes.

6.0.1 Evaluation Metrics

- **Accuracy:** This metric is used to predict the overall percentage of correctly predicted pixels[32].
- **Loss:** This metric tells us how confidently the model made predictions. The lower loss means that the model did a really good job in correctly predicting the pixels based on the damage type[32].

- **F1 Score:** This metric combines precision that shows how many predicted damage pixels were actually damaged and recall that shows how many actual pixels were found. A higher F1 means that the model is accurate and complete. This is a very important metric for classes that are harder to detect, such as severe or collapse, because they are relatively similar[32].
- **IoU (Intersection over Union):** Shows overlap between the predicted damage areas and the ground truth labels[32].
- **Loss Curve:** This is used to plot the change in training and validation loss over time(per epoch). This shows if the model is learning well. Shows signs of underfitting or overfitting and is relatively useful for comparing different models[32].
- **Predicted vs Ground Truth:** This is for visualizing the difference between the ground-truth labels and the predicted masks. This is helpful as it helps to understand the mistakes or patterns that mathematical metrics cannot capture properly.

The table below shows how colors were mapped to different types of damage and their classes for better understanding.

Color	Class	Kind of damage
Black	0	No Damage
Gray	1	Severe Damage
White	2	Collapsed Structures

Table 7: Damage Information

6.1 Baseline U-Net (2-Channel)

6.1.1 Quantitative Results:

- Test Accuracy: 0.8366
- Test Loss: 0.3543
- F1 Score: 0.5303
- IoU: 0.4467
- Pixel Classification Report:

Pixel Classification Report	Precision	Recall	F1-score	Support
Intact	0.89	0.88	0.89	20814694
Severe	0.69	0.71	0.70	7748597
Collapsed	0.00	0.00	0.00	10405
Accuracy			0.84	28573696
Macro avg	0.53	0.53	0.53	28573696
Weighted avg	0.84	0.84	0.84	28573696

Table 8: Classification report for pixel categories including Intact, Severe, and Collapsed.

6.1.2 Qualitative Results:

- Training/Validation Loss Curve [9]: Both training and validation showed a steady decrease over 10 epochs in this case. Training loss decreased from 0.56 to 0.37 while validation loss decreased from 0.45 to 0.35 over Epoch 1 to Epoch 10. Some minor fluctuations were seen in the validation loss from epochs 4 to 7 that indicated some overfitting. However, in general, the loss continued to decrease between the two curves, indicating good learning[32].

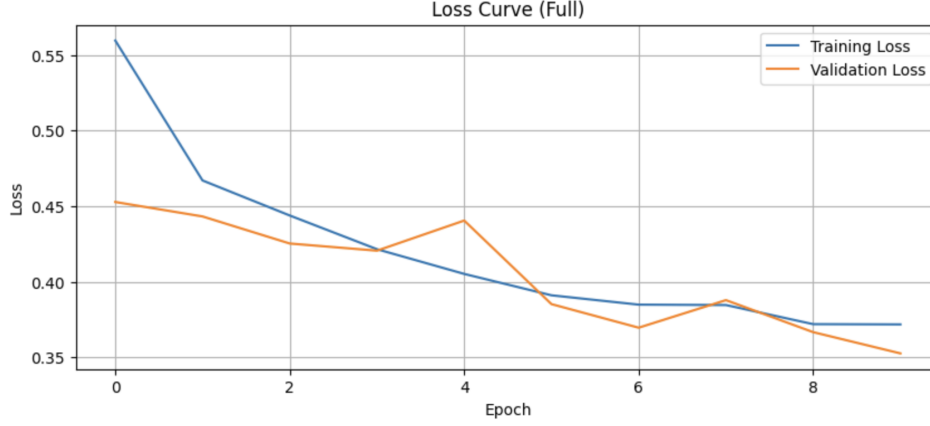


Figure 9: Loss Curve for baseline model

- Prediction vs. Ground Truth: To understand how well the model was performing visually, comparisons were made between the ground truth labels and the models' predicted maps. The ground truth masks were manually labeled in the dataset. For the model predictions, each pixel was assigned an integer code between 0, 1 and 2 depending on the degree of damage. The predicted segmentation maps were generated by the simplified U-Net model, which assigned probabilities in pixels and then converted to the final damage class using the argmax function. The image [10] chosen for visualization is from batch 5 of the test, and it represents the three damage categories.

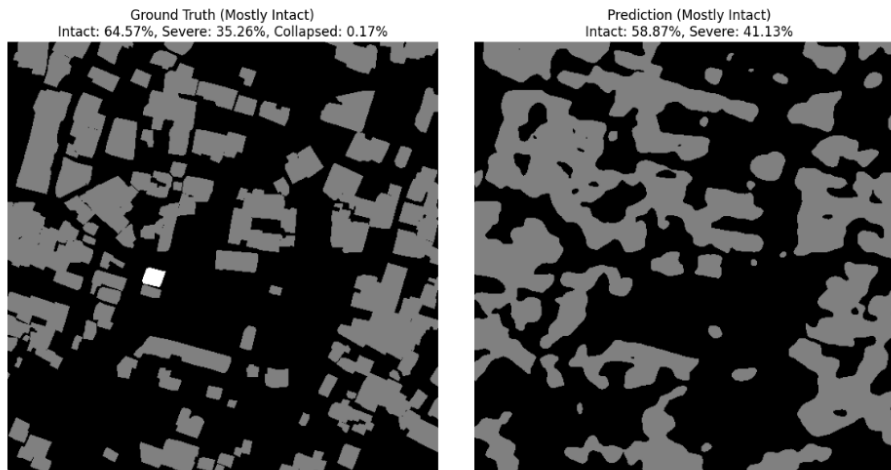


Figure 10: Predicted mask vs Ground-truth label mask for 2-channel Model

According to the ground truth, the pixel distribution in this image was:

- Intact: 64.57%
- Severe: 32.26%
- Collapsed: 0.17%

According to the predicted model, the pixel distribution in this image was:

- Intact: 58.87%
- Severe: 41.13%

Table [7] provides details about the mapping of color to the respective damage class. As seen in the visualized image [10], the ground truth image included all three damages, shown using black, white, and gray colors and predicted. However, the prediction ignored the white region, which represented the collapsed region and only showed gray and black, mostly misclassifying white as gray.

6.2 Basic U-Net (10-Channel)

This model used 10 input channels including the RGB pre-and-post images along with MBI images to give structural information along with 3 other shadow-based images. This was to see if the shadow feature helps to detect damages.

6.2.1 Quantitative Results:

- Test Accuracy: 0.8526
- Test Loss: 0.3452
- F1 Score: 0.54680
- IoU: 0.4682
- Pixel Classification Report :

Category	Precision	Recall	F1-Score	Support
Intact	0.89	0.91	0.90	20122795
Severe	0.77	0.72	0.74	8449283
Collapsed	0.00	0.00	0.00	1618
Accuracy			0.85	28573696
Macro Avg	0.55	0.54	0.55	28573696
Weighted Avg	0.85	0.85	0.85	28573696

Table 9: Pixel Classification Report

6.2.2 Qualitative Results:

- Training/Validation Loss Curve [11]: Here, the loss curve explained how well the model responded to understanding richer input features related to shadows and how well it predicted them. The loss curve plots the training and validation loss of the model over 10 epochs on a log scale. At epoch 0, the training loss is more than 100. But by epoch 1, the training loss

dropped drastically improving the loss and learning well. From Epochs 2 to 9, both training and validation were closely aligned and decreased steadily[32].

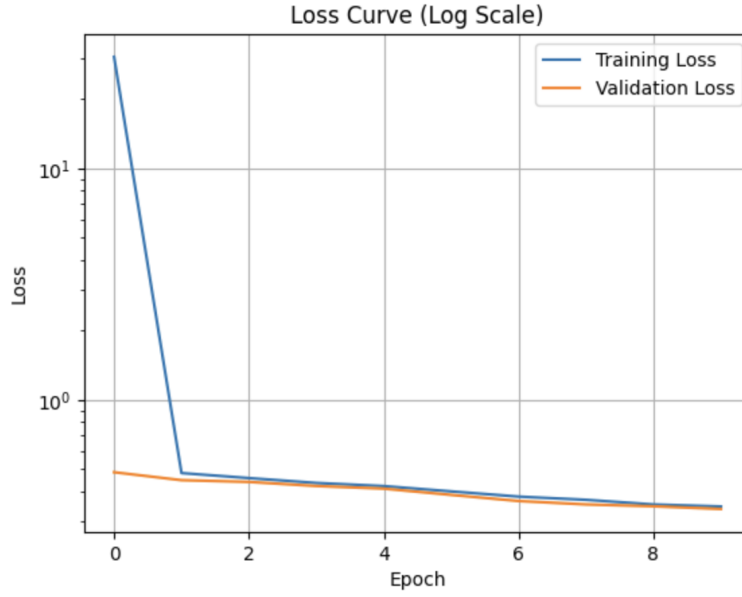


Figure 11: Loss Curve for 10 channel model

- Prediction vs. Ground Truth: The same evaluation process was followed for this model that included skip connections and a deeper architecture, that produced a slightly better segmented output. The image [12] chosen for visualization is from batch 2 of the test and represents all three damage categories. It was the same image [12] that was visualized in the baseline model.

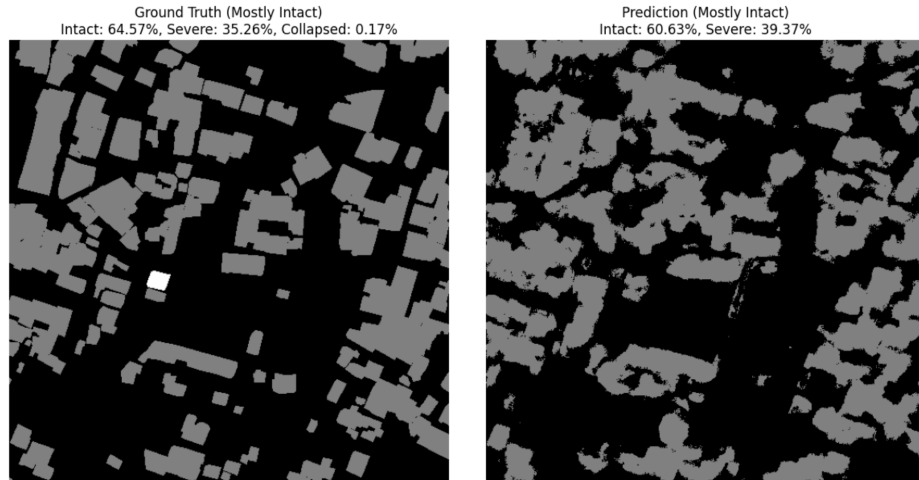


Figure 12: Predicted mask vs Ground-truth label mask for 10-channel Model

According to the ground truth, the pixel distribution in this image was:

- Intact: 64.57%
- Severe: 32.26%
- Collapsed: 0.17%

According to the predicted model, the pixel distribution in this image was:

- Intact: 60.63%
- Severe: 39.37%

So, this model also fails to capture collapsed classes, but also showed a reduced percentage for severe classes and showed an increase in prediction of damage for intact classes from the baseline model. The same color scheme was used as mentioned in Table [refcolormapping]. As seen in the visualized image12, the ground truth image included all three damages, shown using black, white, and gray colors. However, the prediction ignored the white region, which represented the collapsed and only showed gray and black, mostly misclassifying white as gray.

6.3 Comparative Summary of Model Performance

Model	Accuracy (%)	F1 Score (macro)	Mean IoU	Loss
Baseline U-Net (2 ch)	0.8366	0.5303	0.4467	0.3543
Basic U-Net (10 ch)	0.8526	0.54680	0.4682	0.3452

Table 10: Comparative Summary of Model Performance

7 Discussion

7.1 Interpretation of Results

7.1.1 Performance Comparison and Analysis

The results of this research proved that the usage of shadow-related features helped to detect and improve the ability of the deep learning model to detect earthquake-induced damage. After comparing both the Baseline U-Net (2-channel) and the Basic U-Net (10-channel) models, it showed decent improvements across different metrics[32].

The baseline model had 2 inputs consisting of grayscale pre- and post- disaster satellite images. This model was able to predict damaged areas with a test accuracy of 83.66%. It also had an F1 score of 0.5303, IoU of 0.4467, and test loss of 0.3543. However, for this model, the performance was limited, as it was not able to detect collapsed classes at all, and therefore the precision, recall, and F1 score for this damage class was zero[32].

In contrast, the 10 channel U-Net model achieved a higher test accuracy of 85. 26%, a macro F1 score of 0.5468, and a mean IoU of 0.4682. These improvements surely answered the **first research question** showing that shadow features using multichannel inputs were indeed useful for the model to learn and detect damage. The model was even able to identify severe damage (showed a rise in the F1 score from 0.70 to 0.74) much better than the baseline. However, the model was unable to classify and detect collapsed classes at all.

7.1.2 Insights from Loss Curves[32]

The baseline model showed a decent decline in loss, with a minor fluctuation between 4 and 7 epochs. This could have been because of overfitting, meaning that the model was trying to memorize features instead of generalizing due to limited inputs(had only grayscale images).

The 10-channel model started with a very high training loss maybe due to greater complexity of the input fed into it. But, anyhow it later stabilized itself and showed steady improvements with both training and validation loss being closely aligned, indicating good generalization and learning through the rich shadow input features. There was no overfitting here, suggesting that the shadow-based channels helped in improving the model performance and not introducing noise.

7.1.3 Qualitative Observations and Pixel Distribution Analysis

Visual comparisons between predicted masks and ground truth masks further supported the conclusion that shadow features helped in detecting damage. In both models, collapsed class was continuously misclassified and predicted as a severe class, indicating a limitation of class imbalance. This answered the **third research question**, which discussed the limitation of shadow features in deep learning models to detect damage. The amount of severe and intact classes was too high and the collapse class was too low, creating imbalance between classes. The predicted map produced by the 10-channel model had slightly more accurate pixel distribution closer to the ground truth, mainly for the Intact and Severe classes. For example, the image in the result section showed that the 10-channel model had predicted 60.63% Intact and 39.37% Severe compared to the actual baseline model, which only predicted 58.87% Intact and 41.13% Severe. This showed that the 10 channel U-Net model was able to reduce false positives for damage class and predict damage better with more shadow features.

7.1.4 Final Viewpoints

The grayscale pre- and post-images used in the baseline model, captured basic shadow information through reflecting differences in lighting, but was limited in its ability to properly detect and interpret shadows. The initial model was a good starting point for shadow detection. In contrast, the 10-channel U-Net model included many shadow features to be able to achieve more accurate and clear shadow detection.

However, shadow-based features in the 10-channel were able to detect earthquake damage using deep learning models such as U-Net. Therefore, the overall conclusion was able to answer the second **research question** that it is possible for a deep learning model to detect earthquake damages using shadow features in Mexico. But it is important to note that the improvements mentioned in all the metrics were very modest and some damage types like Collapsed were difficult to detect in both the models. This raised a very important point that even though shadows do provide extra information about the structural and geometric changes, their ability to benefit appears to be very dependent on the model architecture, class balance, and quality of the data in the datasets.

The results of this research suggested that shadows can really help in assessing and improving damage detection and should not be ignored. But the benefit is not very dramatic in terms of metrics as of today and requires more detailed research to validate the same. Shadows would be more helpful if they are combined with more advanced models and richer datasets (that are well processed and the classes are well balanced). However, shadow-based U-Net model discussed in this paper is not yet reliable for real-time emergency response in Mexico and to guide long-term good infrastructure planning because of its limitations and thus answering the fourth research question.

7.1.5 Limitations

- **Training time:** One of the main limitations faced in this research was the long training time needed by the standard U-Net model with multichannel input. Each epoch for the 10-channel model took at least three hours to run, making the training process very long and complex. Due to such a long run time, the epoch runs had to be set at 10 which in some way could have restricted the model's full potential.
- **Dataset Imbalance:** The Mexico MEDBFS [1] dataset is very unbalanced. The collapsed class is only 0.047% of the total number of pixels labeled. The intact class is 70.85% and the severe class is 29.09%. This had caused a huge imbalance that made it very difficult for the model to predict the collapse class at all with recall and precision as zero. The model also predicted it wrongly as Severe class.
- **Missing Meta-data :** The data set [1] does not have any meta-data information such as sun angle, building location, and building type. Due to the lack of this information, the scope of this research was limited, as research on region-wise damage was not possible.

7.1.6 Future Work

The results showed good and close to ground-truth results in detecting earthquake building damage using shadow along with satellite images and the U-Net model, but there is always room for more research and development.

- **Better training:** Researchers in the future can use larger batch sizes and more number of training epochs instead of just 10, to check if this improves the model's learning.
- **Better Shadow Handling Method:** Future studies can look at different ways to capture shadows and then detect damage. The method used in this research can be used as a baseline and then developed further with better ideas. An interesting idea could be using Generative Adversarial Networks (GANs) that can help fix lighting issues in regions of tall buildings to detect shadows could be helpful.
- **Meta-Data Availability:** If in future, the data set has meta-data available, then it would be possible to assess damage region-wise, leading to a more powerful research.
- **Other deep learning models:** Future studies can experiment with newer or more advanced deep learning models and compare and improve damage assessment.

8 Conclusion

The purpose of the thesis was to see if it was possible to detect earthquake damage using shadow features when combined with deep learning models. This research compared the baseline model consisting of pre- and post-disaster gray-scale images with a simple U-Net to a 10-channel U-Net model that included multiple shadow inputs.

The research results showed decent improvements in performance metrics such as accuracy, IoU, and F1. Even visual comparisons supported these improvements. The 10-channel U-Net model produced better segmented maps closer to ground truth than the baseline model.

The collapsed class however remained difficult to detect. It was always classified as Severe Class. The results therefore suggested that shadows provide meaningful information that helps in damage

detection but are not enough on their own to detect damage.

However, this research still had some limitations, which need to be worked on in the future, like better training of the model, using more advanced shadow features, using meta-data to predict region-wise damage to be of some help in real-time emergency situations.

Despite these limitations, the research still provided some key insights on how damage assessment can be done using shadows. It also proved that shadows are not just ruins of satellite imagery, but a very important factor in assessing damage when processed properly and combined with deep learning models.

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10 Appendix

The full source code is available on GitHub: https://github.com/sakshicherry4/MexicoEarthquake_thesis